BiteFixes Enterprise Architecture Manual

Volume 02

Engineering Principles & Standards

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Version	Date	Description
1.0	2026	Initial Engineering Principles

Table of Contents

Chapter 1

Introduction

Purpose

This document defines the official engineering standards for the BiteFixes Enterprise Platform.

Every developer, architect and contributor must follow these standards.

These standards ensure:

- Maintainability
- Scalability
- Readability
- Security
- Testability
- Long-term evolution

Chapter 2

Architecture Philosophy

The architecture follows one simple rule.

Business defines software.

Never the opposite.

The implementation order is always:

```
Business

↓

Domains

↓

Use Cases

↓

APIs

↓

Database

↓

Implementation
```

Not:

```
Database

↓

Python

↓

Business
```

Chapter 3

Engineering Principles

Principle 1

Business First

Every feature starts with a business problem.

Never with code.

Principle 2

Domain Driven

Every feature belongs to one domain.

No shared business logic.

Principle 3

Single Responsibility

Each component has one responsibility.

Example

Correct

`CustomerService`

Incorrect

`CustomerTicketInventoryService`

Principle 4

Loose Coupling

Domains communicate through interfaces.

Never direct dependencies.

```
Ticket Domain

↓

ICustomerRepository

↓

Customer Domain
```

Principle 5

High Cohesion

Components that work together stay together.

Principle 6

Configuration Over Code

Everything configurable belongs in configuration.

Examples

- Business Hours
- Supported Languages

- Workflow Steps
- Notification Rules
- AI Parameters

Never hardcoded.

Principle 7

Security by Design

Security is never optional.

Every feature is secure by default.

Principle 8

Observability by Design

Every operation must be observable.

Metrics.

Logs.

Tracing.

Audit.

Principle 9

Event Driven

Important actions generate events.

```
Customer Registered

↓

CustomerRegisteredEvent

↓

Notification

Analytics

Audit

Bitey

Reports
```

Principle 10

AI Independent

AI does not implement business rules.

Business rules belong to the Business Core.

Chapter 4

Domain Driven Design

Each domain owns:

```
Entities
```

Value Objects

Repositories

Policies

Use Cases

Services

Events

DTOs

Validators

Tests

Nothing else.

Chapter 5

Dependency Rule

Dependencies always point inward.

Presentation

↓

Application

↓

Domain

↓

Infrastructure

Never:

Infrastructure

↓

Domain

Chapter 6

Folder Structure

Official project structure

```
app/
    kernel/
    business/
        customer/
        ticket/
        device/
        inventory/
        knowledge/
    ai/
```

```
    integrations/
    infrastructure/
    shared/
    api/
    tests/
docs/
scripts/
```

Every module follows the same organization.

Chapter 7

Naming Standards

Python Files

Correct

```
customer_service.py
```

Incorrect

```
customerServiceFinal.py
```

Classes

CustomerService

TicketRepository

KnowledgeEngine

WorkflowManager

Interfaces

ICustomerRepository

ITicketRepository

IStorageProvider

DTO

CustomerDTO

TicketDTO

DeviceDTO

Events

CustomerRegisteredEvent

TicketClosedEvent

KnowledgePublishedEvent

Use Cases

`RegisterCustomerUseCase`

`CloseTicketUseCase`

`GenerateEstimateUseCase`

Chapter 8

Entity Standards

Every entity must contain

`InternalID`

`UUID`

`CompanyID`

`Status`

`CreatedAt`

`UpdatedAt`

`CreatedBy`

`UpdatedBy`

`Version`

`Metadata`

Mandatory.

Chapter 9

API Standards

Every API must be versioned.

Example

`/api/v1`

`/api/v2`

`/api/v3`

Never break compatibility.

Chapter 10

Database Standards

The database is not the business model.

The database stores business information.

Principles:

- Normalized
- Indexed
- Multi-tenant
- Soft Delete
- Audit Ready
- UUID Public IDs

- BIGINT Internal IDs

Chapter 11

Security Standards

Authentication

Authorization

Permissions

Roles

Encryption

Secrets

Audit Logs

JWT

Rate Limiting

HTTPS

Chapter 12

Logging Standards

Every operation logs:

`Timestamp`

User

Company

Module

Action

Execution Time

Result

Error

Chapter 13

Observability

Monitor:

API Response Time

Database Performance

AI Latency

Queue Size

Memory Usage

CPU Usage

Error Rate

Ticket Volume

Chapter 14

Testing

Every module must include

```
Unit Tests

Integration Tests

API Tests

Security Tests

Performance Tests
```

Chapter 15

AI Standards

Bitey follows:

```
Perception

↓

Understanding

↓

Memory

↓
```

```
Intent

↓

Knowledge

↓

Planning

↓

Decision

↓

Execution

↓

Reflection

↓

Learning
```

AI never manipulates the database directly.

Chapter 16

Development Workflow

```
Business Requirement

↓
```

Architecture Review

↓

Domain Design

↓

Use Cases

↓

API

↓

Database

↓

Implementation

↓

Tests

↓

Documentation

↓

Deployment

Chapter 17

Architecture Review Checklist

Before implementing any feature answer:

✓ Which domain owns it?

✓ What business problem does it solve?

✓ Which events are generated?

✓ Which policies apply?

✓ Which APIs change?

✓ Which database objects change?

✓ Which AI capability uses it?

✓ Which tests validate it?

Chapter 18

Golden Rule

Every line of code must answer:

Does this improve the business model without breaking the architecture?

If the answer is **No**, redesign before implementing.

Conclusion

This document establishes the engineering foundation for BiteFixes Enterprise Platform.

Following these principles ensures that the platform remains:

- Maintainable
- Scalable
- Modular
- Secure
- Observable
- AI-Ready
- Enterprise-Grade